

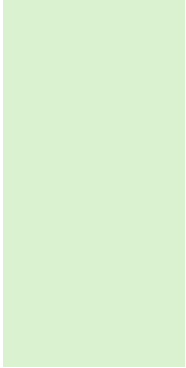
key	menu	T	T_lead	feature1...	unit_cost	unit_price	qty_stock
A001	100	-2	0	1			
A001	100	-2	0	0			
A001	100	-2	0	1			
A003	100	-2	0	1			
A003	100	-2	0				
A005	234	-1	0	0			
A005	234	-1	0	1			
A005	234	-1	0	0			
A005	234	-1	0	1			
A001	0	0	0	1			
A001	0	0	0	0			
A001	0	0	0	1			
A003	0	0	0	1			
A003	0	0	0	1			
A005	0	0	0	0			
A005	0	0	0	1			
A005	0	0	0	1			
A005	0	0	0	0			
A006							

qty_outstanding_T+1	...featureN	qty	historically_available	historically_chosen
		1	1	1
		2	0	
		3	0	
		1	0	
		1	0	
		4	1	1
			1	
		6		
		1	completion	
		2		
		3		
		1		
		2		
		1		
		2		
		3		
		4		

profit

\$3

(\$4)



Cell: A1

Note: key column:

Any kind of asset/ticker identifier: SKU, contract_id, user_id, etc.

Cell: B1

Note: menu column:

Identifier of a "menu": a full list of possible trade options at any given decision moment. One date may have multiple menus - meaning multiple portfolios could have been assembled at that date

Cell: C1

Note: T - time column:

time column is either:

0 - identifying current task to come up with predictions;

a negative number - the past from current time (a week number, minute, etc.);

a positive number - for simplicity of time tracking we allow absolute time IDs (e.g. unix timestamp)

Cell: D1

Note: T_lead column [optional]:

Lead time in same T units. Lead time is how much time typically passes before any sale may happen. This helps to better reference records from "Sales" table. Can be None, empty, or absent.

Cell: E1

Note: feature columns:

Any number of features or naïve predictors

Cell: F1

Note: Special optional feature columns: cost

cost is a special type of a column that will be interpreted by the system

Cell: G1

Note: Special optional feature columns: price:

price is a special column type that will be treated by the system as current or historical price. If more historical prices are present it is advisable to add them as unrolled prices: price_T-1, price_T-2, etc., back in history.

Columns with naming pattern price1_T-1, price2_T-1 - are different possible price measures, if relevant

Cell: H1

Note: Special optional feature columns: stock_qty column (*stock* wildcard series of columns)

Columns representing currently open position (status sellable) for this key, or amount of this item currently in stock. We assume a single market for all keys, so exiting the position is done on FIFO (first in - first out) basis if there were multiple orders in the past. Profit depends on whether the sales could be realized of previous stock (full position close) and only then current new QTY. If no sales of this batch happened, profit will be negative (full cost write-off)

Cell: I1

Note: Special optional feature columns: qty_outstanding_T+1 column:

represents all inventory that must be considered in FIFO process, but not currently in sellable status. T+1 suffix represents that this outstanding amount is expected to arrive in next 1 time unit. T+0 is technically impossible, and must be included in qty_stock

Cell: K1

Note: qty column:

qty is a deal size column. Deal size can be integer, count-type values (like 1 apple, 2 apples) or floating point value (like \$151.50 or 38.566kg), but not both and not a combination.

Cell: L1

Note: historically_available column:

Binary column (values only 1 or 0) that represent whether this particular option was available at decision time. For example, the features and target outcome are known, but the option (e.g. quantity) was not present as available trade option at decision time

Cell: M1

Note: historically_available column:

Binary column (values only 1 or 0) that represent whether this particular option was chosen as part of previous business operations at decision time. For example, the features and target outcome may be known, but the option (e.g. quantity) was not actually traded by the business. This defines model's risk appetite and other business process calibrations.

Cell: N1

Note: profit column: a total profit of decision/deal

This column defines the known, or unknown target outcome per trade. If the outcome is precisely known, the profit column must have a value. If precise value is unknown, then this value must be blank (or None), and instead any approximate values must be present in the features. It is important to propagate approximate values for all rows in this case.

Profit column is optional as internal replay module may recreate profits from Sales table

Cell: C11

Note: Rows with T=0: inference-time (predict)

Rows with T=0 indicate the "now" menu that needs to be predicted, the values in "profit" will be filled in (predicted) to yield a portfolio-sum risk-calibrated result